Robust Joint Motion Forecasting Under Missing and Noisy Observations

Compact MotionLM-style joint forecaster on the Waymo Open Motion Dataset — corruption-aware training and bounded RL post-training

Status: PRELIMINARY (development set, 3 seeds; RL pending). Development numbers use manifests/dev.json; held-out numbers use the locked manifests/final.json (final evaluation pending).

Research question and motivation

Question

Does a joint two-agent autoregressive forecaster keep its accuracy and physical consistency when the observed history is incomplete or noisy — and can training on corrupted observations, plus bounded reward post-training, make it more robust without hurting clean accuracy?

Why it matters

Deployed trackers drop frames and jitter. Forecasters trained only on clean logs can degrade sharply; robustness must be measured with the same evaluator on the same scenes, with uncertainty, before claiming an improvement.

Approach

Reproduce a compact MotionLM-style joint model (M0), train a matched corruption-aware variant (M1) and an independent-agent ablation, then post-train M1 with REINFORCE + KL + replay against simpler controls. Preregistered thresholds; paired group bootstrap.

- Preregistered success rule (PLAN.md): ≥ 5 % relative reduction of core-suite joint minADE@6, ≤ 3 % clean worsening, ≤ 0.5 pp overlap increase
- Scope kept to forecasting, robustness and bounded RL — no observation selection, no architecture expansion

Task and dataset

9,144
training windows (9,144 after removing scenes that share a source segment with dev)

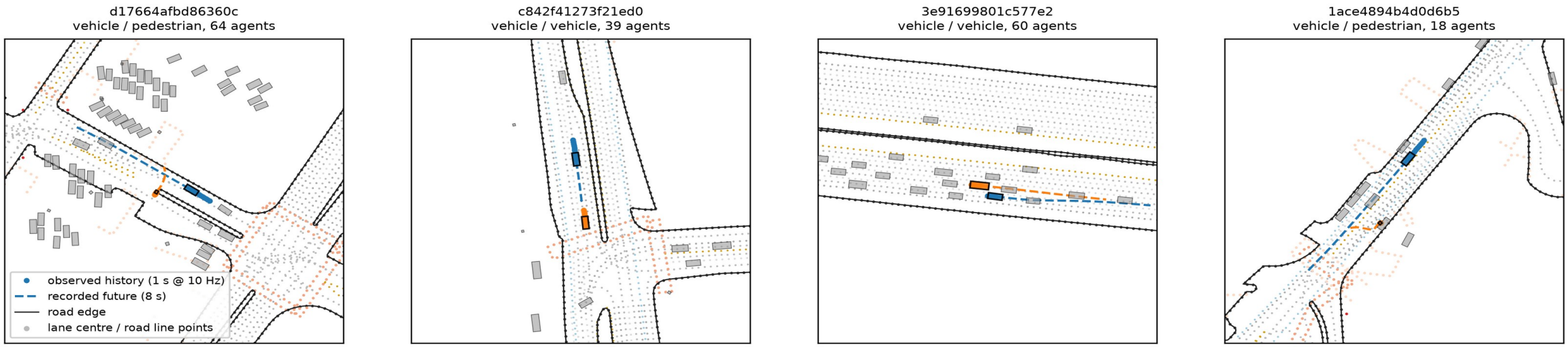
500
development scenes (selection, tuning, sampling freeze)

1,000
final held-out scenes, LOCKED (official validation shards)

1 s → 8 s
history @ 10 Hz → 16 joint steps @ 2 Hz

2
interacting targets per scene (objects_of_interest)

Waymo Open Motion Dataset v1.3.0 — interactive pairs (blue, orange) with map and context agents (grey)

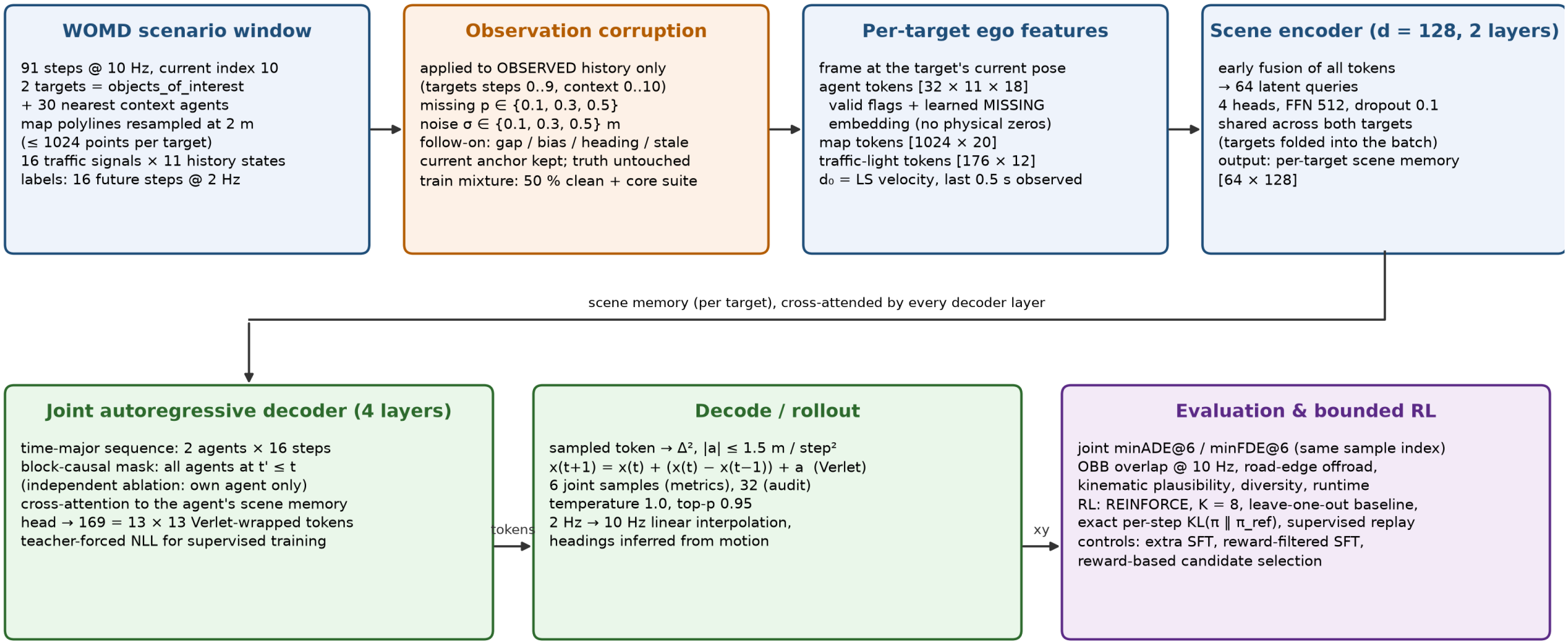


- WOMD v1.3.0 scenario format, 48 training + 12 validation shards; 1.7 % of pair scenarios excluded by the 10 Hz timing rule (dropped frames, not interpolated)
- Calibration on dev: tokenizer saturation 0.08 %, reconstruction 0.092 m; offroad convention validated by log playback (0.98 % offroad); logged-future overlap reference 1 % on jointly valid intervals (audit v2)

PRELIMINARY

Implemented architecture

Compact MotionLM-style joint forecaster — 1.79 M parameters (as implemented in src/waymo_av/model.py)



PRELIMINARY

Experimental design and baselines

Method	Training observations	Role
Constant position / constant velocity	— (deterministic, 1 sample)	non-learned floor; CV uses the model's own d_0 estimator
Independent decoder (Mind)	clean	ablation: same model, no cross-agent attention
Joint M0	clean	clean-trained baseline (reference for all paired deltas)
Joint M1	50 % clean + core corruption suite	corruption-aware model
M1 + extra SFT	as M1, matched extra steps	control: more supervised training
M1 + reward-filtered SFT	own samples, reward-filtered	control: reward signal without policy gradient
M1 + bounded RL	REINFORCE, K = 8, LOO baseline, KL, replay	post-training (reward: overlap, offroad, kinematics)

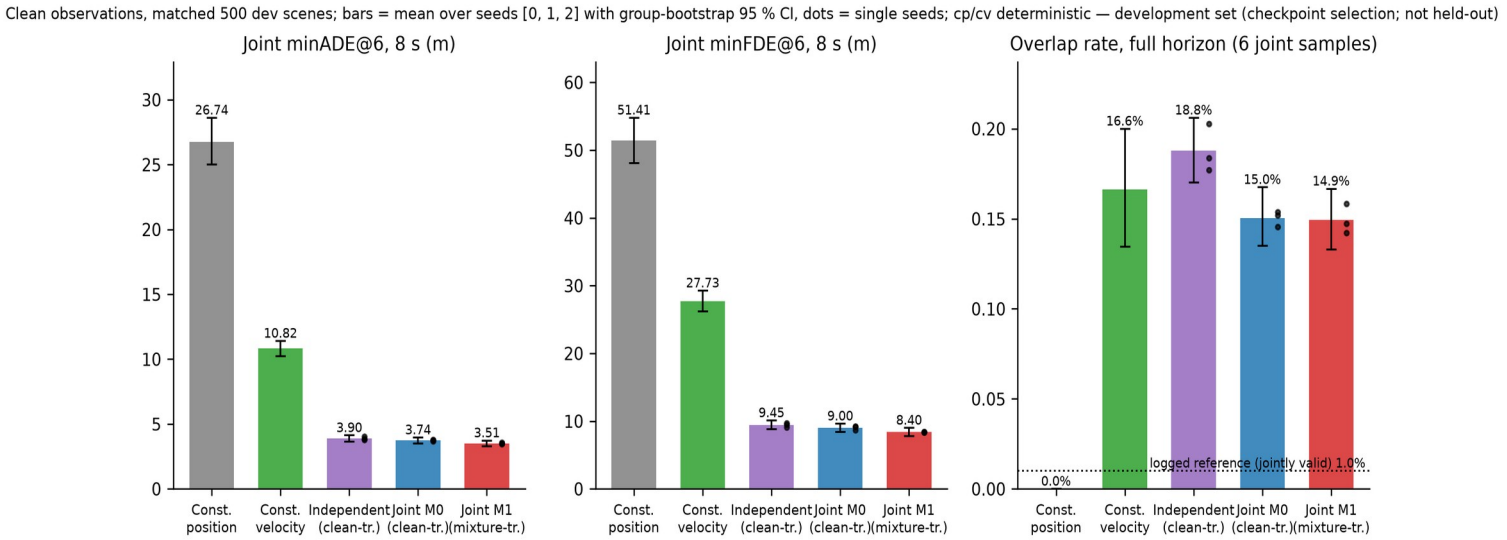
Corruption suite (observations only)

Core: missing 10/30/50 % of history frames; Gaussian position noise σ = 0.1/0.3/0.5 m.
Held-out severities: missing 70 %, σ = 1.0 m, compound.
The current anchor frame is kept; ground truth is never corrupted.

Measurement

Joint minADE@6 / minFDE@6 over the same sample index (6 samples; 32-sample audit for overlap/diversity).
OBB overlap at 10 Hz, road-edge offroad (vehicles), kinematic plausibility, diversity.
Paired bootstrap over heuristic source-segment groups; 3 seeds; checkpoints chosen on dev only; final set evaluated once after freeze.

Clean forecasting results



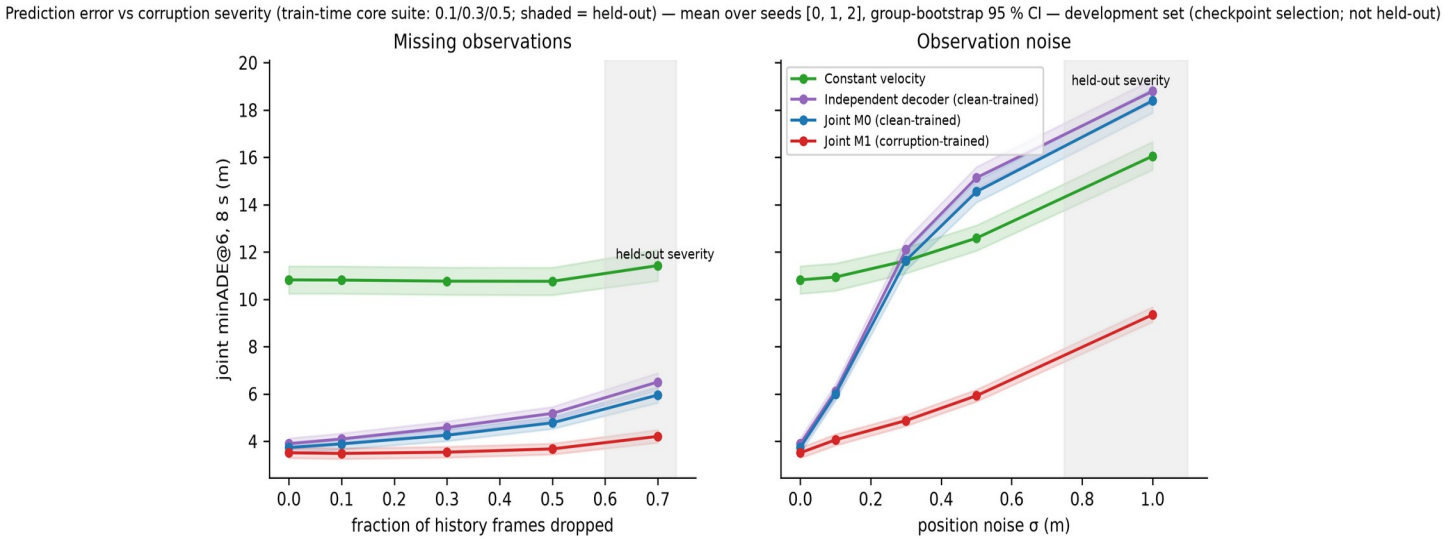
Joint minADE@6 (m), clean, seeds 0,1,2

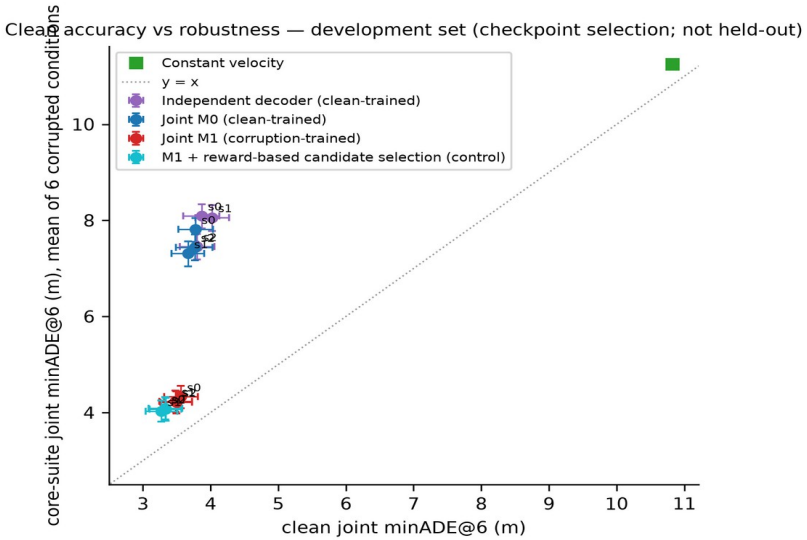
Constant velocity: 10.82 [10.24, 11.40]
Independent: 3.90 [3.66, 4.13]
Joint M0: 3.74 [3.50, 3.96]
Joint M1: 3.51 [3.29, 3.73]
Joint − independent: -0.16 [-0.28, -0.05] (-4.1 %)

- Evaluated on the development set (500 scenes) — PRELIMINARY; identical scenes and sampling (6 joint samples, T = 1.0, top-p 0.95) for every learned model; cp/cv are deterministic
- Overlap, clean, 32-sample audit: M0 15.3 [13.8, 16.9] % over the full generated horizon; on jointly valid intervals M0 15.2 [13.7, 16.8] % vs logged reference 1.0 [0.2, 2.0] % (same approximate test applied to logs; not a lower bound)
- Checkpoints selected on dev by their own criterion (M0: clean jminADE; M1: mean of clean and suite); 12k steps, 1.79 M parameters, d = 128

PRELIMINARY

Robustness to missing and noisy observations





seed	core-suite rel. Δ (≤ −5 %)	clean rel. Δ (≤ +3 %)	core overlap Δ pp (≤ +0.5)	verdict
0	-44.6 % [-47.3, -41.9]	-5.7 % [-10.3, -0.9]	+0.63 [-0.66, +1.94]	FAIL
1	-42.1 % [-44.8, -39.3]	-5.1 % [-9.7, -0.4]	+0.92 [-0.47, +2.26]	FAIL
2	-43.3 % [-46.1, -40.4]	-7.2 % [-12.5, -1.8]	-0.81 [-2.08, +0.54]	PASS

M1 vs M0 (paired, group bootstrap)

Core suite jminADE: -3.26 [-3.42, -3.09] (-43.3 %)
Held-out suite: -5.25 [-5.49, -4.99] (-45.5 %)
Clean: -0.22 [-0.33, -0.11] (-6.0 %)
Core overlap: +0.25 [-0.67, +1.20] pp

PRELIMINARY

Bounded RL post-training and controls

PENDING — figure not yet generated

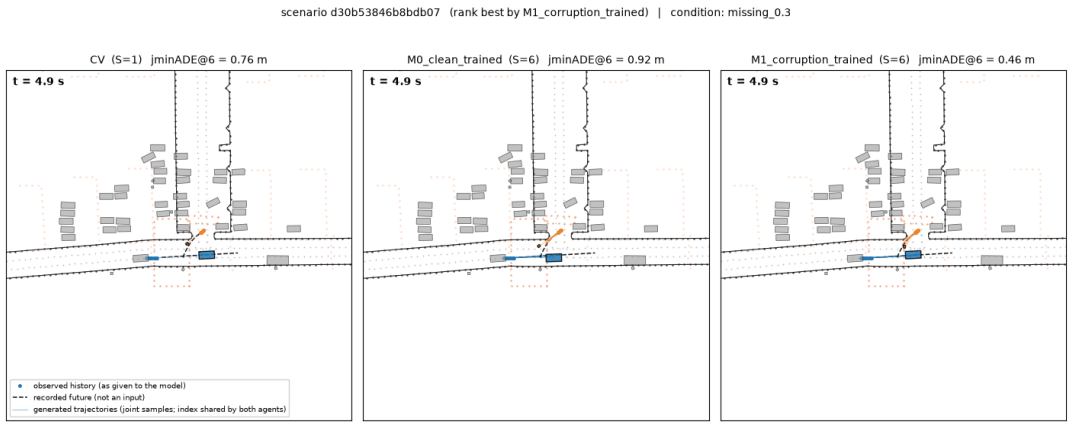
RL vs its M1 initialisation (clean, paired)

PENDING — post-training stage not yet run. Runs planned: rl_pilot, filtered_sft_pilot, extra_sft_pilot per seed, all initialised from the dev-selected M1 checkpoint.

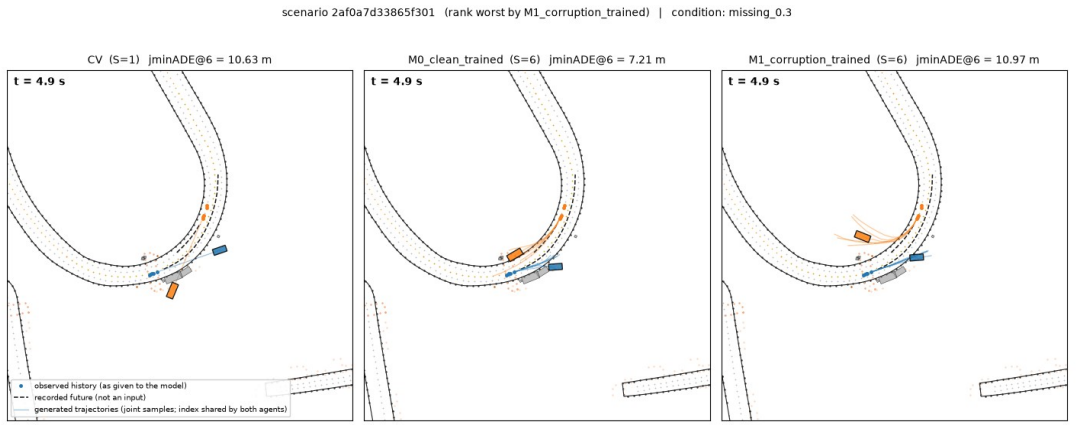
Controls vs M1 (overlap, 32 samples)

PENDING — controls run in the same stage. Interpretation rule (frozen before final evaluation): an RL effect counts only if it exceeds both the extra-SFT and reward-filtered-SFT controls under identical sampling.

Qualitative examples and failures



Success case — scenario d30b53846b8bdb07, condition missing_0.3 — CV 0.76 m, M0_clean_trained 0.92 m, M1_corruption_trained 0.46 m



Failure case — scenario 2af0a7d33865f301, condition missing_0.3 — CV 10.63 m, M0_clean_trained 7.21 m, M1_corruption_trained 10.97 m

- Selection rule (not hand-picked): pool = first 64 manifest scenes; select 'mixed' by M1_corruption_trained joint minADE@6; identical scenes/corruption/sampling across methods
- Each panel shows the identical scene, corruption realisation and sampling; dots = observed history as given to the model, black dashed = recorded future (never an input), coloured = generated joint samples with one shared sample index for both agents
- Animations (GIF + MP4, synchronized, fixed camera): reports/demos/<set>/<scenario>_<condition>.{gif,mp4}

PRELIMINARY (development set, 3 seeds; RL pending)

Conclusions, limitations, remaining work

- Clean: joint M0 3.74 [3.50, 3.96] m vs CV 10.82 [10.24, 11.40] m; joint − independent -0.16 [-0.28, -0.05] (-4.1 %)
- Robustness: M1 − M0 on the core suite -3.26 [-3.42, -3.09] (-43.3 %); clean -0.22 [-0.33, -0.11] (-6.0 %); verdicts: seed 0 FAIL, seed 1 FAIL, seed 2 PASS
- RL and controls: pending
- Limitations: compact model and ~9k training scenes (not a leaderboard reproduction); official metrics package unavailable (custom metrics audited on oracles); the 2 Hz box-overlap test is approximate (logged reference 1 % on jointly valid intervals; RL reward = approximate geometric consistency, not verified safety); source-segment grouping is heuristic; synthetic corruptions are stress tests, not sensor models
- Remaining: RL + controls (running), freeze method choices, single locked final evaluation, final report and deck